1️⃣ What is a Route Table?

A Route Table controls **where traffic goes.**

NSG = controls **who is allowed**

Route Table = controls **where traffic is sent**

It works at **Layer 3 (Routing level)**.

Route tables

2️⃣ Default Azure System Routes

Every subnet automatically has system routes:

Address Prefix	Next Hop
VNet CIDR	Virtual Network
0.0.0.0/0	Internet
RFC1918	None
AzureLoadBalancer	Load Balancer

You don't see them unless you check **Effective Routes**.

3️⃣ What is UDR (User Defined Route)?

When you create your own route table → it overrides system routes.

Used for:

- Force traffic to Firewall
- Force traffic to NVA
- Block Internet
- Hub-Spoke architecture
- On-prem routing via VPN/ExpressRoute

4️⃣ Route Table Components

Field	Meaning
Address Prefix	Destination CIDR
Next Hop Type	Where traffic goes
Next Hop Address	IP (if Virtual Appliance)

5️⃣ Next Hop Types (Very Important for Exam)

Type	Meaning
Internet	Send to public internet
Virtual Network	Stay inside VNet
Virtual Appliance	Send to Firewall/NVA
VNet Peering	Send to peered VNet
None	Drop traffic
Virtual Network Gateway	Send to VPN/ER

🔥 LAB 1 – Block Internet Using Route Table

🎯 Scenario:

Block subnet from accessing Internet using Route Table (not NSG).

Step 1 – Create Route Table

Portal → Search → Route Tables → Create

- Name: RT-NoInternet
- Region: Same as VNet

Step 2 – Add Route

Add route:

Setting	Value
Address Prefix	0.0.0.0/0
Next Hop Type	None

This means:

“All internet-bound traffic → Drop”

Step 3 – Associate Route Table

Go to:

Route Table → Subnets → Associate

Select:

Your Subnet

Step 4 – Test

Inside VM:

```nginx
ping google.com
```

Should fail.

🔥 LAB 2 – Force Traffic to Firewall (NVA Scenario)

🎯 Scenario:

All traffic must go through firewall VM (10.0.0.4).

Add route:

Address Prefix	0.0.0.0/0
Next Hop Type	Virtual Appliance
Next Hop IP	10.0.0.4

Associate to subnet.

Now all outbound traffic goes to firewall.